

Pair-Chain Transfer Operators and Random Synchronization

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This paper (Paper XXVI of the RIME program) develops a stochastic pair-transfer line for synchronizing automata. It separates deterministic pair control, global reset control, pairwise random absorption, and global random synchronization.

Abstract

Problem. Deterministic pair control, global reset control, pairwise random absorption, and global random synchronization are distinct quantities, and their asymptotic scales need not agree.

Approach. We retain the labelled unordered-pair graph exactly: its integer adjacency counts the letters that preserve a distinct pair, and normalization produces a substochastic transfer operator. The fundamental matrix describes expected absorption, while Perron–Frobenius theory describes the dominant tail scale.

Results. For the standard Černý family, we recover the known uniform-input worst-pair mean in pair-transfer coordinates,

$$H_2(C_n) = n^3 - \frac{3}{2}n^2 + \frac{n \bmod 2}{2}$$

and derive the Perron asymptotic

$$1 - \rho(Q_{C_n}) = \frac{\pi^2}{8n^3}(1 + o(1)).$$

Thus its worst-pair mean is cubic although its deterministic reset threshold is quadratic. An extremal advance/reset family gives

$$rt(R_n) = n - 1, \quad H_2(R_n) = E[\tau_{\text{sync}}] = 2^n - 2, \quad 1 - \rho(Q_{R_n}) \sim 2^{-n}.$$

Finally, for an alphabet of size k and reset threshold r , we prove the sharp universal waiting envelope

$$H_2(A) \leq E[\tau_{\text{sync}}] \leq \frac{k^{r+1} - k}{k - 1}.$$

The bound is sharp for every $k \geq 2$: a generalized advance/reset family attains equality. Equality requires two simultaneous extremal features: a shortest reset word of maximal self-overlap, necessarily a constant-letter word a^r , and a pair that does not absorb before the first occurrence of that pattern. The comparison separates optimal deterministic control, pairwise control, and random-word absorption. In particular, polynomial deterministic synchronization does not imply polynomial stochastic absorption without additional structural hypotheses.

Boundary. Exact Bernoulli-input worst-pair formulas for the Černý family and the qualitative existence of exponentially slow random synchronization families are prior results. The paper does not claim a general polynomial relation between reset threshold and random absorption, a spectral proof of the deterministic Černý conjecture, or a nontrivial zero theory for the finite determinant function introduced below. Appendix A supplies the perturbative control used by

the leading Perron result and proves a further second-order refinement; it is not used by TA-II or TA-III.

Keywords: synchronizing automata; pair automaton; transfer operator; Perron–Frobenius theory; absorbing Markov chains; pattern waiting time; Černý automata.

Notation

Symbol	Meaning
$A = (Q, \Sigma, \delta)$	complete deterministic finite automaton
$n = Q , k = \Sigma $	state and alphabet sizes
$rt(A)$	shortest reset-word length
$d_2(x, y)$	shortest word merging the pair $\{x, y\}$
$D_2(A)$	$\max_{x \neq y} d_2(x, y)$
$\mathcal{P}_2(Q)$	unordered distinct pairs of states
$\widehat{T}_A$	transient pair multigraph adjacency
$Q_A = \widehat{T}_A/k$	uniform random-letter transient matrix
τ_p	random absorption time from pair p
$H_2(A)$	worst-pair mean $\max_p E[\tau_p]$
τ_{sync}	first time all pairs have merged under one common random word
$E[\tau_{\text{sync}}]$	mean global random synchronization time
$\rho(Q_A)$	Perron root of the transient matrix

1 Introduction

A deterministic synchronizing automaton admits a word that sends all states to one state. The classical extremal quantity is the reset threshold $rt(A)$, the minimum length of such a word. It is an optimal-control quantity: at each step the controller chooses the next letter [3, 18].

There is a different natural experiment. Draw letters independently and uniformly from Σ and ask how long a given pair takes to merge. This is not optimal control. It is absorption in a finite substochastic Markov chain. The two experiments share the same automaton but need not share a complexity scale.

The exact carrier of the random experiment is the unordered-pair transfer operator:

$$A \mapsto \widehat{T}_A \mapsto Q_A \mapsto ((I - Q_A)^{-1}, \rho(Q_A)).$$

This representation should be compared with, but not identified with, the deterministic transformation-semigroup and mass-quotient representations:

$$\text{one automaton} \longrightarrow \begin{cases} \text{controlled words and rank descent,} \\ \text{random letters and pair absorption.} \end{cases}$$

The deterministic pair-hitting, marked-kernel, and Schreier viewpoints used for comparison here were developed in the previous deterministic study [5]. That work studies controlled rank descent, whereas this paper studies random absorption for a fixed automaton.

The distinction is forced by the examples considered here. The Černý family has quadratic optimal reset and cubic random absorption. The advance/reset family has linear optimal reset and exponential random absorption. Deterministic and stochastic scales are therefore not monotone across families. The qualitative separation and the exact Černý mean have prior antecedents; Section 8 states the attribution boundary precisely.

1.1 Contributions

The paper organizes its results into one exact representation and three theorem groups.

1. The pair-chain transfer operator is exact: it is the integer adjacency of a labelled multi-graph, not a norm-based coarse graining.
2. **TA-I (diffusive Černý regime)**. The known uniform-input worst-pair mean is recovered in pair-transfer coordinates, and the PF gap is shown to have leading constant $\pi^2/8$ at scale n^{-3} together with a second-order refinement.
3. **TA-II (extremal rare-run regime)**. An advance/reset family attains the exact value $2^n - 2$ for both worst-pair and global random synchronization means, and its PF gap is asymptotic to 2^{-n} .
4. **TA-III (sharp reset-word envelope)**. For alphabet size k and reset threshold r , both the worst-pair mean and the global random synchronization mean are bounded by $(k^{r+1} - k)/(k - 1)$, and this envelope is sharp for every $k \geq 2$.

1.2 Claim boundary

This paper does not prove a deterministic reset bound from a spectral gap. It does not identify $D_2(A)$ with $rt(A)$, and it does not infer random absorption from a deterministic mass potential. It also does not claim the known exact Černý mean or the qualitative existence of exponential random-input examples as new results.

The leading Černý spectral argument is part of the main theorem surface. Appendix A supplies its uniform perturbation and root-window estimates and then continues to the second-order refinement. The refinement is not needed for the elementary TA-II/TA-III results.

2 The exact pair-chain representation

Let

$$\mathcal{P}_2(Q) = \{\{x, y\} : x, y \in Q, x \neq y\}.$$

For $p = \{x, y\}$ and $a \in \Sigma$, either $a(p)$ is another distinct pair or $a(x) = a(y)$ and the pair has merged. Define

$$\widehat{T}_A[p, q] = \#\{a \in \Sigma : a(p) = q\},$$

and the merge multiplicity

$$m_A(p) = \#\{a \in \Sigma : a(x) = a(y)\}.$$

Every row satisfies the exact conservation law

$$\sum_q \widehat{T}_A[p, q] + m_A(p) = k.$$

Consequently

$$Q_A = \frac{1}{k} \widehat{T}_A$$

is substochastic. Missing row mass is precisely one-step absorption probability.

Proposition 2.1 (exactness).

$\hat{T}_A$ is the source-row adjacency matrix of the transient unordered-pair multigraph, with one edge for every letter that preserves distinctness. No numerical quotient or norm aggregation enters its definition.

Proof. A deterministic letter maps each pair to exactly one image. If the image is distinct it contributes one to exactly one matrix entry; otherwise it contributes one to $m_A(p)$. Summing over letters gives the row identity.

Proposition 2.2 (synchronization and transience).

For a finite complete deterministic automaton, the following are equivalent:

1. A is synchronizing;
2. every distinct pair is mergeable by some word;
3. every state of the pair chain can reach absorption;
4. $\rho(Q_A) < 1$.

Proof. A reset word merges every pair, so (1) implies (2). If every pair is mergeable, repeatedly choose a pair in the current image and append a word merging it; the image size strictly decreases, proving (2) implies (1). Statements (2) and (3) are the same reachability assertion in the pair graph. For a finite substochastic matrix, every state reaches missing row mass if and only if there is no closed stochastic communicating class, which is equivalent to spectral radius strictly below one.

Proposition 2.3 (fundamental matrix).

Assume A is synchronizing. The expectation vector $h_p = E_p[\tau]$ is

$$h = (I - Q_A)^{-1} \mathbf{1} = \sum_{j \geq 0} Q_A^j \mathbf{1}.$$

Proof. First-step decomposition gives $h = \mathbf{1} + Q_A h$. Proposition 2.2 makes $I - Q_A$ invertible and the Neumann series convergent.

For a single random input stream, couple all initial pairs with the same word and define the global random synchronization time

$$\tau_{\text{sync}} = \max_{p \in \mathcal{P}_2(Q)} \tau_p.$$

It follows that

$$H_2(A) = \max_p E_p[\tau] \leq E[\tau_{\text{sync}}].$$

The inequality can be strict because expectation does not commute with the maximum.

Proposition 2.4 (PF hitting sandwich).

Suppose Q_A is irreducible and let $v > 0$ be a right Perron vector. Then

$$\frac{1}{1 - \rho(Q_A)} \leq H_2(A) \leq \frac{v_{\max}}{v_{\min}} \frac{1}{1 - \rho(Q_A)}.$$

Proof. The fundamental matrix $N = (I - Q_A)^{-1}$ is nonnegative and has spectral radius $(1 - \rho)^{-1}$. Its infinity norm is the maximum row sum, which equals $H_2(A)$, proving the lower bound. Since $\mathbf{1} \leq v/v_{\min}$,

$$N\mathbf{1} \leq \frac{Nv}{v_{\min}} = \frac{v}{v_{\min}(1 - \rho)},$$

which gives the upper bound.

The eigenvector ratio is essential. Reversibility can make spectral analysis easier, but does not by itself bound $v_{\max}/v_{\min}$ uniformly.

3 Four inequivalent synchronization scales

The pair-chain representation makes four scales visible:

$$D_2(A) = \max_{x \neq y} d_2(x, y), \quad rt(A), \quad H_2(A), \quad E[\tau_{\text{sync}}].$$

$D_2(A)$ is optimal control for one pair. $rt(A)$ is optimal simultaneous control of the full state set. $H_2(A)$ is the worst-pair mean under a fixed random policy, while $E[\tau_{\text{sync}}]$ is the mean time for all pairs to merge under the same random word.

Every reset word merges every pair, hence

$$D_2(A) \leq rt(A).$$

The inequality is generally strict. For the Černý family,

$$D_2(C_n) = \frac{n(n-1)}{2}, \quad rt(C_n) = (n-1)^2.$$

No corresponding general polynomial comparison with $H_2(A)$ or $E[\tau_{\text{sync}}]$ exists; Section 5 gives an exact counterexample.

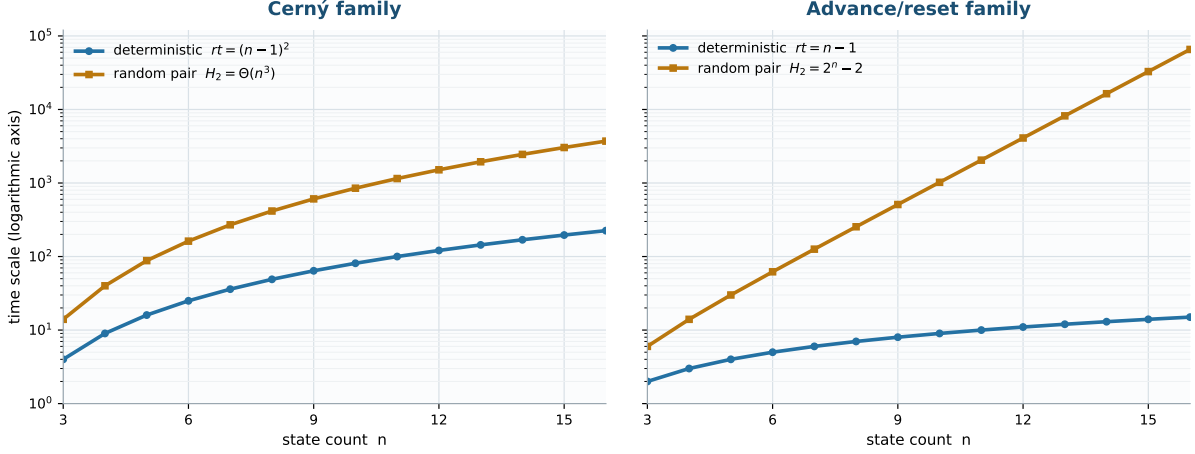
The two families studied below also reverse the deterministic ordering. For all sufficiently large n ,

$$rt(R_n) = n-1 < rt(C_n) = (n-1)^2, \quad H_2(R_n) = 2^n - 2 > H_2(C_n).$$

Thus the family with the shorter deterministic reset threshold can have the larger random pair-absorption scale. Any bridge between these carriers must retain information about the distribution and overlap structure of useful words; deterministic reset complexity alone cannot supply it.

Deterministic and Random Synchronization Scales Separate

Shortest-word control is min-plus; pair absorption averages the labelled transition carrier



The family with the shorter reset threshold can have the larger random absorption time.

Figure 1. The Černý and advance/reset families reverse their ordering between deterministic reset threshold and worst-pair random absorption. The former has quadratic reset and cubic absorption scales; the latter has linear reset and exponential absorption scales.

4 The diffusive Černý pair chain

Let C_n have state set $\mathbb{Z}/n\mathbb{Z}$ and two letters:

$$a(x) = x + 1 \pmod{n},$$

and

$$b(n-1) = 0, \quad b(x) = x \quad (x \neq n-1).$$

The deterministic identity $rt(C_n) = (n-1)^2$ is classical.

Lemma 4.1 (phase reduction).

Let $H(i, j) = E_{\{i, j\}}[\tau]$ for $0 \leq i < j \leq n-1$, and let $g(d) = H(0, d)$. While $j < n-1$, the letter b is a self-loop and a moves both coordinates forward, so first-step decomposition gives

$$H(i, j) = 2 + H(i+1, j+1).$$

Solving this recurrence together with the boundary equations at $j = n-1$ gives the exact phase formula

$$H(i, j) = g(j-i) - 2i.$$

In particular $H(i, j) \leq g(j-i)$, so a worst pair can always be chosen with left endpoint 0.

Proof. The displayed first-step equation follows from $H = 1 + \frac{1}{2}H + \frac{1}{2}H \circ a$. At the defect boundary, a sends $\{i, n-1\}$ to $\{0, i+1\}$ and b sends it to $\{0, i\}$; substituting $H(i, j) = g(j-i) - 2i$ into these boundary equations reduces them exactly to the recurrence for g below. Uniqueness of the finite absorbing-chain system proves the formula.

Theorem 4.2 (Gusev's uniform-input worst-pair mean, recovered).

For $n \geq 3$,

$$H_2(C_n) = n^3 - \frac{3}{2}n^2 + \frac{n \bmod 2}{2}.$$

This is the uniform-input specialization of Gusev's Bernoulli-input worst-pair formula [11]. The proof below recovers it directly in the pair-transfer coordinates used here.

Proof. By Lemma 4.1, it remains to compute $g(d) = E_{\{0,d\}}[\tau]$, $1 \leq d \leq n-1$; no rotation symmetry of the full pair chain is being assumed. Eliminating the deterministic interior rotations between visits to the defect gives

$$g(d) = 2(n-1-d) + 1 + \frac{g(n-d) + g(n-1-d)}{2}, \quad 1 \leq d \leq n-2,$$

with

$$g(n-1) = 1 + \frac{g(1)}{2}.$$

Direct substitution verifies the unique solution

$$g(d) = 2d(2n(n-1) - (2n-1)d).$$

This concave quadratic is maximized at the central integer distance. Evaluating at that distance gives the displayed parity-sensitive formula.

Lemma 4.3 (boundary generating-function collapse).

For $0 \leq x < x+d \leq n-1$, let

$$F_{x,d}(z) = E_{\{x,x+d\}}[z^\tau], \quad G_d(z) = F_{0,d}(z).$$

If $x < n-1-d$, first-step decomposition gives

$$F_{x,d}(z) = \frac{z}{2}F_{x,d}(z) + \frac{z}{2}F_{x+1,d}(z),$$

and hence, with $c_d(z) = (z/(2-z))^{n-1-d}$,

$$G_d = \frac{z}{2}c_d(G_{n-d} + G_{n-1-d}) \quad (1 \leq d \leq n-2),$$

while

$$G_{n-1} = \frac{z}{2}(G_1 + 1).$$

Let $C_n(z) = \text{diag}(c_1(z), \dots, c_{n-1}(z))$ and define the symmetric $(n-1) \times (n-1)$ matrix R_0 by

$$R_0[d, n-d] = 1 \quad (1 \leq d \leq n-1),$$

and

$$R_0[d, n-1-d] = 1 \quad (1 \leq d \leq n-2),$$

with all other entries zero. Then the boundary system is

$$\left(I - \frac{z}{2}C_n(z)R_0 \right) G(z) = \frac{z}{2}e_{n-1}.$$

Proof. In the interior, b fixes the pair and a advances both coordinates. Iterating the displayed interior identity reaches the unique boundary pair containing $n-1$. At that boundary, a gives the $n-d$ term, b gives the $n-1-d$ term, and for $d = n-1$ the latter transition is absorption.

Lemma 4.4 (exact base spectrum).

Put $\theta = \pi/(2n-1)$. The eigenpairs of $R_0/2$ are

$$\lambda_k = (-1)^{k+1} \cos(k\theta), \quad u_k(d) = \sin(2kd\theta), \quad 1 \leq k, d \leq n-1.$$

In particular the base Perron eigenvalue and a positive Perron vector are

$$\rho_0 = \cos \theta, \quad u_1(d) = \sin(2d\theta) > 0.$$

Proof. For $d < n-1$, the two nonzero entries in row d give

$$(R_0 u_k)(d) = u_k(n-d) + u_k(n-1-d) = 2(-1)^{k+1} \cos(k\theta) u_k(d).$$

The last row gives the same identity using $u_k(1) = 2(-1)^{k+1} \cos(k\theta) u_k(n-1)$. The sine vectors are mutually orthogonal by the finite sine identity

$$\sum_{d=1}^{n-1} \sin(2kd\theta) \sin(2ld\theta) = \frac{2n-1}{4} \delta_{k\ell}.$$

They are therefore nonzero and form an eigenbasis, giving the full spectrum. Positivity identifies $k=1$ as the Perron mode.

Lemma 4.5 (Perron pole and pencil root).

Define

$$S_n(z) = \frac{z}{2} C_n(z)^{1/2} R_0 C_n(z)^{1/2}.$$

If $z_* > 1$ is the first positive solution of $\lambda_{\max}(S_n(z)) = 1$, then

$$z_* = \rho(Q_{C_n})^{-1}.$$

Proof. The transient pair graph of C_n is irreducible and has a self-loop, hence Q_{C_n} is primitive. The positive left and right Perron vectors imply that the Perron pole has nonzero residue in every pair-hitting generating function, whose common radius is therefore $\rho(Q_{C_n})^{-1}$. Lemma 4.3 is an exact elimination of those functions. Moreover, Since $C_n(z)$ is positive diagonal for $1 < z < 2$, there is an actual similarity

$$C_n(z)^{-1/2} \left(\frac{z}{2} C_n(z) R_0 \right) C_n(z)^{1/2} = \frac{z}{2} C_n(z)^{1/2} R_0 C_n(z)^{1/2} = S_n(z).$$

Hence $I - (z/2)C_n(z)R_0$ is similar to $I - S_n(z)$. For $1 < z < 2$, the matrix $S_n(z)$ is nonnegative and entrywise increasing; therefore its Perron eigenvalue is the first eigenvalue to reach one. This identifies its first positive root with the Perron pole.

Theorem 4.6 (Černý PF asymptotic).

Let $\rho_n = \rho(Q_{C_n})$. Then

$$1 - \rho_n = \frac{\pi^2}{8n^3} (1 + o(1)).$$

Consequently

$$(1 - \rho_n) H_2(C_n) \rightarrow \frac{\pi^2}{8}.$$

Proof. By Lemmas 4.3 and 4.5, $z_* = 1 + \varepsilon_*$ is characterized by

$$\lambda_{\max}(S_n(1 + \varepsilon_*)) = 1, \quad \rho_n = z_*^{-1}.$$

Lemma 4.4 identifies the simple base Perron eigenvalue $\rho_0 = \cos(\pi/(2n-1))$. In its normalized sine eigenbasis, the uniform perturbation estimate proved in Appendix A gives, for $\varepsilon = O(n^{-3})$,

$$\lambda_{\max}(S_n(1 + \varepsilon)) = \rho_0 + \mu_0 \varepsilon + O(n^2 \varepsilon^2), \quad \mu_0 = \rho_0 \left(n - \frac{1}{2} \right) + O(n^{-3}).$$

The root-window argument in Appendix A first shows $\varepsilon_* = O(n^{-3})$, so this estimate may be evaluated at the root. Writing $\delta = 1 - \rho_0 = \Theta(n^{-2})$ gives

$$\delta = \mu_0 \varepsilon_* + O(n^2 \varepsilon_*^2), \quad \varepsilon_* = \frac{\delta}{\mu_0} (1 + o(1)).$$

Since $1 - \rho_n = \varepsilon_*/(1 + \varepsilon_*)$, it follows that

$$1 - \rho_n = \frac{1 - \rho_0}{\rho_0(n - 1/2)} (1 + o(1)) = \frac{\pi^2}{8n^3} (1 + o(1)).$$

The product limit follows from Theorem 4.2.

4.1 Second-order refinement

A sharper analytic refinement is given by Theorem A.1. It identifies the coefficient $\gamma_2 = \pi^2/24 - 3/4$ for the normalized product $(1 - \rho_n)H_2(C_n)$ and includes the uniform third-order remainder, the parity bookkeeping, and the root-window argument. The leading controls in Appendix A are required by Theorem 4.6; the additional second-order conclusion is not used by TA-II or TA-III.

5 The rare-run regime

Let R_n have states $\{0, \dots, n-1\}$, put $z = n-1$, and define

$$a(i) = \begin{cases} i+1, & i < z, \\ z, & i = z, \end{cases} \quad b(i) = \begin{cases} 0, & i < z, \\ z, & i = z. \end{cases}$$

Theorem 5.1 (linear reset, exponential absorption).

For $n \geq 2$,

$$\boxed{rt(R_n) = n-1, \quad H_2(R_n) = E[\tau_{\text{sync}}] = 2^n - 2.}$$

Proof. The word a^{n-1} sends every state to z , so $rt(R_n) \leq n-1$. The pair $\{0, z\}$ cannot merge before the first coordinate has advanced $n-1$ times without an intervening b , giving the reverse inequality.

Under random letters, z remains fixed. Starting from $\{0, z\}$, letter a advances the first coordinate and letter b resets it to zero. Absorption is therefore the waiting time for a run of $n-1$ consecutive a 's. The standard run recurrence gives

$$E[T_r] = 2 + 2^2 + \dots + 2^r = 2^{r+1} - 2.$$

With $r = n-1$, this is $2^n - 2$. For $i > 0$, the pair $\{i, z\}$ starts with partial progress toward the same run and has expectation $2^n - 2^{i+1} < 2^n - 2$. A pair not initially containing z either merges

at the next b , or enters one of these z -pair states after a run of a 's; its first-step recurrence is therefore also bounded by the $\{0, z\}$ value. The whole automaton also synchronizes exactly at the first run of $n - 1$ copies of a : state 0 cannot reach the fixed state z sooner, whereas such a run sends every state to z . Hence the worst-pair and global means coincide.

Theorem 5.2 (rare-run PF gap).

Let $\rho_n = \rho(Q_{R_n})$. Then

$$2(1 - \rho_n)(2\rho_n)^{n-1} = 1$$

and hence

$$1 - \rho_n = 2^{-n}(1 + O(n2^{-n})).$$

In particular,

$$(1 - \rho_n)H_2(R_n) \rightarrow 1.$$

Proof. Ordering transient pairs by whether they contain z , the full matrix is block triangular. The block of pairs not containing z is nilpotent, while the z -pair block is irreducible and carries the Perron root. This Perron block consists of $\{0, z\}, \dots, \{n - 2, z\}$. With $v_0 = 1$, its right eigenvector recurrence is

$$v_{j+1} = 2\rho_n v_j - 1,$$

and therefore

$$v_j = (2\rho_n)^j - \frac{(2\rho_n)^j - 1}{2\rho_n - 1}.$$

The absorbing boundary $v_{n-1} = 0$ gives the characteristic equation. Writing $\varepsilon_n = 1 - \rho_n$ in that equation gives

$$\varepsilon_n = 2^{-n}(1 - \varepsilon_n)^{-(n-1)} = 2^{-n}(1 + O(n2^{-n})).$$

The product limit follows from Theorem 5.1.

Corollary 5.3 (no unconditional polynomial bridge).

There is no universal implication from a polynomial reset threshold to either random-input mean:

$$rt(A_n) = \text{poly}(n) \not\Rightarrow \begin{cases} H_2(A_n) = \text{poly}(n), \\ E[\tau_{\text{sync}}] = \text{poly}(n). \end{cases}$$

and no universal polynomial lower bound on $1 - \rho(Q_A)$ can follow from a polynomial reset threshold alone.

6 A sharp reset-word waiting envelope

For a word w of length r , let $B(w) \subseteq \{1, \dots, r\}$ be its border lengths, including r itself. For an iid uniform k -letter stream, the exact pattern waiting formula is

$$E[T_w] = \sum_{j \in B(w)} k^j.$$

Theorem 6.1 (universal envelope).

Let A be a synchronizing automaton with at least two states over an alphabet of size $k \geq 2$, and let $rt(A) = r \geq 1$. Then

$$H_2(A) \leq E[\tau_{\text{sync}}] \leq \frac{k^{r+1} - k}{k - 1}.$$

Proof. Fix a shortest reset word w of length r . Whenever w occurs as a contiguous block in the random stream, every currently surviving pair is merged by the end of that block. Thus $\tau_p \leq T_w$ under the natural coupling, for every initial pair p . The border formula and $B(w) \subseteq \{1, \dots, r\}$ give

$$E[\tau_p] \leq E[T_w] \leq k + k^2 + \dots + k^r = \frac{k^{r+1} - k}{k - 1}.$$

The first inequality was established after Proposition 2.3. The occurrence of w merges all pairs under the same coupling, so $\tau_{\text{sync}} \leq T_w$ almost surely. Taking expectations gives the second inequality.

Proposition 6.2 (sharpness for every alphabet size).

For every $k \geq 2$ and $r \geq 1$, there exists a synchronizing automaton with alphabet size k , reset threshold r , and

$$H_2(A) = E[\tau_{\text{sync}}] = \frac{k^{r+1} - k}{k - 1}.$$

Proof. Use states $\{0, \dots, r\}$ with sink $z = r$. Let one letter a advance $i \mapsto i + 1$ until z , and let each of the other $k - 1$ letters reset every non-sink state to 0 while fixing z . Then a^r is a shortest reset word. The pair $\{0, z\}$ and the full state set synchronize exactly when the random stream first contains r consecutive copies of a . For success probability $1/k$, the classical run-waiting expectation is

$$E[T_{a^r}] = k + k^2 + \dots + k^r = \frac{k^{r+1} - k}{k - 1}.$$

Thus both inequalities in Theorem 6.1 are sharp for every $k \geq 2$.

Proposition 6.3 (equality mechanism).

Let A have alphabet size k and reset threshold r . The pairwise endpoint $H_2(A)$ equals the upper envelope in Theorem 6.1 if and only if there exist a letter a and a pair p such that a^r is a shortest reset word and, under the uniform random input,

$$\tau_p = T_{a^r} \quad \text{almost surely.}$$

Equivalently, the extremal case combines maximal reset-word autocorrelation with no premature absorption of the extremal pair.

Proof. If these conditions hold, a^r has every border length $1, \dots, r$, so its waiting time attains the upper bound in Theorem 6.1, and the pair attains the same expectation. Conversely, suppose equality holds and choose a pair p attaining $H_2(A)$. For any shortest reset word w of length r , the natural coupling gives $\tau_p \leq T_w$ almost surely. Hence equality of the universal bound forces equality throughout

$$E[\tau_p] \leq E[T_w] \leq \sum_{j=1}^r k^j.$$

The second equality requires all border lengths $1, \dots, r$, which forces $w = a^r$ for some letter a . Since $T_w - \tau_p \geq 0$ and has expectation zero, $T_w = \tau_p$ almost surely.

The envelope is exponential in reset threshold and therefore cannot support a general polynomial spectral shortcut. The Černý family lies far below this worst envelope, whereas the generalized rare-run construction saturates it. This separates diffusive pair transport from pattern-autocorrelation slowness.

7 Discussion: determinant encoding

For any finite transfer matrix T , one may define

$$Z_T(s) = \det(I - sT)^{-1}.$$

This is a rational function. It is analytic near $s = 0$ and has automatic meromorphic continuation to the Riemann sphere. Its finite poles are the reciprocals of the nonzero eigenvalues of T , counted with algebraic multiplicity. Because its numerator is the constant one, it has no finite zeros.

Equivalently, the reciprocal polynomial

$$L_T(s) = \det(I - sT)$$

has zeros at the reciprocal eigenvalues. This repackages finite spectral data; it does not by itself create a nontrivial analytic-continuation or critical- line problem. For the pair chain, the meaningful dominant singularity is $s = \rho(Q_A)^{-1}$, because it controls absorption asymptotics.

Accordingly, the proof-facing objects are Q_A , its resolvent, its Perron eigenpair, and explicit family geometry. The determinant notation is only a compact encoding of finite spectral data, not a separate zeta theory.

8 Related Work and Novelty Boundary

8.1 Deterministic synchronization

The deterministic line asks for one controlled word or one admissible rank- descent route. Its algebra is existential and min-plus. The pair-chain line averages all letters under a fixed random policy. Its algebra is linear and Perron–Frobenius.

The rare-run theorem shows that a bridge cannot depend only on $rt(A)$ or on a deterministic potential that merely bounds $rt(A)$. A valid future bridge must add information controlling the distribution of useful words. Plausible hypotheses include:

- bounded reset-word border complexity or autocorrelation;
- a Doeblin or minorization condition on rank-decreasing events [14];
- reversible pair chains with a controlled Perron eigenvector ratio [13];
- typed per-rank coupling between random pair motion and deterministic checkpoint geometry.

None of these is asserted generally here. The current conclusion is a clean separation of representations:

same automaton, different carriers, different complexity notions.

8.2 Random-input and transfer-operator literature

The pair graph is classical in deterministic synchronization theory, where it supports pair-merging distances and reset-word arguments [18, 7, 1]. The recent deterministic treatment [5] further retains marked transformation kernels and Schreier-corridor geometry. The present use is different: every letter-preserving pair transition is retained with its integer multiplicity and then normalized to a substochastic matrix. The resulting fundamental matrix and Perron root belong to the random-input problem, not to the optimal-control problem.

Gusev derived exact worst-pair mean formulas for the Černý family under Bernoulli input and showed that the selected central pair is maximal [11]. Setting both letter probabilities to $1/2$ yields Theorem 4.2. Gusev’s cycle and defect letters are named oppositely to ours, so any comparison of the nonuniform formulas must swap the associated probability labels. Theorem 4.2 is therefore a recovery in the present transfer notation, not a new expectation formula.

The same work also exhibited a fixed automaton family with exponential random synchronization time under a linearly long prescribed pattern [11]. Related studies treat random automaton models and random-input synchronization under their respective hypotheses [17, 2, 4]. Accordingly, TA-II does not claim the qualitative deterministic–stochastic separation as new. Its contribution is the extremal advance/reset realization with exact mean $2^n - 2$, its Perron-gap asymptotic, and its role in attaining the sharp universal envelope and equality mechanism of Section 6.

The reset-word envelope uses classical border and overlap formulas for waiting for one prescribed word [8, 9, 10]. A full pair chain can absorb through many words, so the single-pattern formula is an upper-bound device rather than a complete description of its spectrum. Absorbing-chain and quasi-stationary arguments used here are standard [12, 6, 15, 16].

The references listed below are the release bibliography. The comparison is bounded to the cited literature and is not an exhaustive novelty claim.

9 Claim status and evidence

The paper keeps proof, exact finite calculation, and bounded computation separate. The following table is the reader-facing claim map.

Claim	Status	Carrier or source
Exact pair-chain representation and absorbing-chain equations	Theorem	Sections 2–3
Known uniform-input Černý expectation, recovered	Theorem	Theorem 4.2; Gusev [11]
TA-I leading Perron-gap asymptotic	Theorem	Theorem 4.6
TA-II extremal rare-run formulas and Perron gap	Theorem	Theorems 5.1–5.2
TA-III universal waiting envelope	Theorem	Theorem 6.1
TA-III sharpness and equality mechanism	Theorem / Proposition	Propositions 6.2–6.3
Second-order Černý refinement	Theorem	Theorem A.1

The numerical scripts validate exact constructions, reproduce the displayed finite observations, or cross-check asymptotic formulas. They do not upgrade a computational observation to a theorem. The public evidence companion is in `experiments/paper26/`.

10 Conclusion and outlook

The unordered-pair transfer operator is an exact stochastic carrier for a fixed deterministic automaton. It makes four scales explicit: one-pair control, simultaneous reset, pairwise random absorption, and global random synchronization. The Černý and rare-run families show that these scales can separate, while the reset-word envelope identifies the sharp pattern-waiting obstruction. In particular, the two families reverse their ordering between deterministic reset threshold and worst-pair random absorption.

The new results concern the Černý pair-chain Perron asymptotics and second-order refinement, the exact extremal rare-run realization, and the sharp reset-word envelope with its equality mechanism. They do not form a bridge from random spectra to the deterministic Černý conjecture. A future bridge would need hypotheses controlling useful-word distribution or the geometry of the Perron eigenvector. The deterministic mass/macrograph program and the stochastic pair-chain program may share automata, but they do not share a theorem interface automatically.

The finite determinant function $Z_T(s) = \det(I - sT)^{-1}$ remains useful as a compact encoding of finite spectral data. Its rationality and meromorphic continuation are automatic; any deeper zero statement would be a different problem and is outside the scope of this paper.

Appendix A Explicit Černý second-order refinement

The proof below records the exact pencil, selection-rule estimates, uniform third-order bound, and root expansion needed for the second-order coefficient. The paper-owned companion contains a supplementary line-by-line verification of the explicit constants; that record is not a logical premise of the theorem.

Theorem Appendix A.1 (Second-order Černý refinement). *As $n \rightarrow \infty$,*

$$1 - \rho_n = \frac{\pi^2}{8n^3} \left[1 + \frac{3}{2n} + \left(\frac{3}{2} + \frac{\pi^2}{24} \right) \frac{1}{n^2} + o(n^{-2}) \right].$$

Consequently,

$$(1 - \rho_n)H_2(C_n) = \frac{\pi^2}{8} \left[1 + \left(\frac{\pi^2}{24} - \frac{3}{4} \right) \frac{1}{n^2} + o(n^{-2}) \right].$$

In particular, the relative n^{-1} coefficient vanishes and

$$\gamma_1 = 0, \quad \gamma_2 = \frac{\pi^2}{24} - \frac{3}{4}.$$

The proof occupies Sections A.1–A.4.

A.1 Uniform Feshbach estimate

Let $\theta = \pi/(2n-1)$, $\rho_0 = \cos \theta$, and write the symmetrized two-channel pencil in the eigenbasis of the unperturbed operator as $H(\varepsilon) = a(\varepsilon)\Lambda + b(\varepsilon)\widetilde{D}$, where

$$\begin{aligned} f_m(\varepsilon) &= \frac{1+\varepsilon}{2} \left(\frac{1+\varepsilon}{1-\varepsilon} \right)^m, \\ A &= f_{(n-2)/2}, \quad B = f_{(n-1)/2}, \quad a = A + B, \quad b = \frac{B-A}{2}. \end{aligned}$$

For completeness, split the matrix of Lemma 4.3 as $R_0 = R_a + R_b$, where $R_a[d, n-d] = 1$ and $R_b[d, n-1-d] = 1$ on their respective index ranges, and put $D = R_b - R_a$. If U is the orthonormal sine eigenbasis from Lemma 4.4, then

$$\Lambda = U^T(R_0/2)U = \text{diag}(\lambda_1, \dots, \lambda_{n-1}), \quad \widetilde{D} = U^T D U.$$

These definitions give the displayed two-channel pencil exactly. They also fix the later notation $g_k = \rho_0 - \lambda_k$ and $v = (\widetilde{D}_{21}, \dots, \widetilde{D}_{n-1,1})^T$.

For the audited constants $N_3 = 64$ and $c_0 = 4$, and for $n \geq N_3$ and $0 \leq \varepsilon \leq c_0 n^{-3}$, Taylor's theorem gives

$$\begin{aligned} a &= 1 + a_1 \varepsilon + a_2 \varepsilon^2 + r_a, & |r_a| &\leq 8n^3 \varepsilon^3, \\ b &= b_1 \varepsilon + b_2 \varepsilon^2 + r_b, & |r_b| &\leq 8n^2 \varepsilon^3, \end{aligned}$$

with

$$a_1 = n - \frac{1}{2}, \quad a_2 = \frac{2n^2 - 2n - 1}{4}, \quad b_1 = \frac{1}{4}, \quad b_2 = \frac{2n - 1}{8}.$$

The derivative bounds behind these constants are uniform on the whole window: $f_m \leq 0.501$, $|L'_m| \leq 1.001n$, $|L''_m| \leq 2$, and $|L'''_m| \leq 2.01n$, where $L_m = \log f_m$. For the odd channel, writing $q = ((1 + \varepsilon)/(1 - \varepsilon))^{1/2}$ and $b = A(q - 1)/2$ gives $|b'''| \leq 48n^2$.

A.2 Lower-block control

The mode gaps satisfy

$$g_k = \rho_0 - \lambda_k, \quad g_k \geq 4n^{-2} \quad (k \geq 2),$$

using the even/odd parity split of the cosine modes. If $G = \text{diag}(g_k)_{k \geq 2}$ and v is the Perron-row coupling, the selection-rule sums can be made explicit as follows.

Lemma Appendix A.2 (Perron-row selection rule). *For $2 \leq k \leq n - 1$,*

$$\boxed{\widetilde{D}_{1k} = (-1)^k \frac{4 \sin \theta}{2n - 1} \frac{\sin(k\theta)}{\cos \theta - (-1)^k \cos(k\theta)}}.$$

Moreover, for even k ,

$$|\widetilde{D}_{1k}| \leq \frac{8\pi^2}{3(2n - 1)k}, \quad g_k = \cos \theta + \cos(k\theta) \geq \frac{1}{2},$$

whereas for odd $k \geq 3$,

$$|\widetilde{D}_{1k}| \leq \frac{8\pi^2 k}{(2n - 1)^3}, \quad g_k = \cos \theta - \cos(k\theta) \geq \frac{16k^2 \theta^2}{9\pi^2}.$$

Proof. By the finite sine orthogonality identity in Lemma 4.4, the unnormalized vectors satisfy $\|u_k\|^2 = (2n - 1)/4$. Evaluating $u_1^T D u_k$ with the finite Dirichlet sine sum gives the boxed identity. For even k , factor $\cos \theta - \cos(k\theta)$ into two sines; for odd k , use $\cos \theta + \cos(k\theta) \geq \cos \theta$. The elementary bounds $2x/\pi \leq \sin x \leq x$ on $[0, \pi/2]$ give the two displayed estimates. Summing the even and odd classes separately, using $\sum_{k \geq 1} k^{-2} = \pi^2/6$, gives

$$\begin{aligned} v^T G^{-1} v &\leq \frac{16\pi^6}{27(2n - 1)^2} + \frac{18\pi^4}{(2n - 1)^3}, \\ v^T G^{-2} v &\leq \frac{1}{(2n - 1)^2} \left(\frac{32\pi^6}{27} + \frac{81\pi^6}{32} \right). \end{aligned}$$

In particular, for $n \geq 64$,

$$v^T G^{-1} v \leq 200n^{-2}, \quad v^T G^{-2} v \leq 1000n^{-2}.$$

Finally, $\|\widetilde{D}\| = \|D\| \leq 2$ by orthogonal invariance.

Relative to the Perron mode, the exact block decomposition is

$$H(\varepsilon) = \begin{pmatrix} a\rho_0 + b\widetilde{D}_{11} & bv^T \\ bv & a\Lambda_{\perp} + b\widetilde{D}_{\perp} \end{pmatrix}.$$

For an eigenvalue λ in the Perron branch, define the lower Schur block

$$\boxed{K(\lambda, \varepsilon) = \lambda I - (a\Lambda_{\perp} + b\widetilde{D}_{\perp}).}$$

The diagonal entry obeys $|\widetilde{D}_{11}| \leq 3n^{-3}$. Weyl's inequality and the scalar bounds imply, throughout the same window,

$$E = (a-1)I + (\lambda - a\rho_0)G^{-1} - bG^{-1/2}\widetilde{D}_{\perp}G^{-1/2},$$

and hence

$$\|E\| \leq 2n\varepsilon + n^2\varepsilon \leq \frac{8}{n^2} + \frac{4}{n} < \frac{1}{2},$$

for the weighted lower-block perturbation $K(\lambda, \varepsilon) = G^{1/2}(I + E)G^{1/2}$. Thus K is invertible before the Perron root is specialized, and

$$|v^T(K^{-1} - G^{-1})v| \leq 800\varepsilon.$$

A.3 Remainder and coefficient assembly

Taking the Schur complement of the displayed block gives the exact identity

$$\lambda = a\rho_0 + b\widetilde{D}_{11} + b^2 v^T K^{-1} v.$$

Consequently

$$\lambda = \rho_0 + \mu_0\varepsilon + \nu_0\varepsilon^2 + R_3(n, \varepsilon),$$

where

$$\mu_0 = a_1\rho_0 + b_1\widetilde{D}_{11}, \quad \nu_0 = a_2\rho_0 + b_2\widetilde{D}_{11} + b_1^2 v^T G^{-1} v,$$

and

$$\boxed{|R_3(n, \varepsilon)| \leq 16n^3\varepsilon^3.}$$

The four contributions are bounded respectively by $8n^3\varepsilon^3$, $24\varepsilon^3/n$, $800\varepsilon^3$, and $2\varepsilon^3$. The last bound uses

$$|b - b_1\varepsilon| \leq \left(\frac{n}{4} + \frac{32}{n}\right)\varepsilon^2, \quad |b + b_1\varepsilon| \leq \frac{3}{4}\varepsilon.$$

A.4 Root window and the coefficient

The same estimates give $\mu_0 \geq 0.95n$ and $\nu_0 > 0$. Put

$$\varepsilon_b = \frac{2(1 - \rho_0)}{\mu_0}.$$

Then

$$\varepsilon_b < 4n^{-3}, \quad |R_3(n, \varepsilon_b)| < \frac{1 - \rho_0}{2}.$$

Since $\lambda_{\max}(H(\varepsilon_b)) > 1$ and the original symmetrized matrix $S_n(z)$ is entrywise nonnegative with every nonzero entry strictly increasing for $z \in (1, 2)$, its Perron eigenvalue is strictly increasing. Hence its unique root lies in the stated window. At that root, put $\delta = 1 - \rho_0$. The expansions used in the coefficient assembly are

$$\mu_0 = n - \frac{1}{2} - \frac{\pi^2}{8n} - \frac{\pi^2}{16n^2} + O(n^{-3}), \quad \nu_0 = \frac{n^2}{2} + O(n),$$

and the root equation gives

$$\varepsilon_* = \frac{\delta}{\mu_0} - \frac{\nu_0 \delta^2}{\mu_0^3} + O(n^{-7}).$$

Substitution, followed by the elementary expansion of ρ_0 , yields

$$1 - \rho_n = \frac{\pi^2}{8n^3} \left[1 + \frac{3}{2n} + \frac{3/2 + \pi^2/24}{n^2} + o(n^{-2}) \right],$$

and, using Theorem 4.2,

$$(1 - \rho_n)H_2(C_n) = \frac{\pi^2}{8} \left[1 + \frac{\pi^2/24 - 3/4}{n^2} + o(n^{-2}) \right].$$

The relative n^{-2} coefficient separates into

$$\gamma_2^{\text{linear}} = \frac{5\pi^2}{48} - \frac{3}{4}, \quad \Delta\gamma_2^{\text{curvature}} = -\frac{\pi^2}{16}.$$

Therefore

$$\gamma_2 = \frac{\pi^2}{24} - \frac{3}{4}.$$

The proof is uniform for $n \geq 64$, which is sufficient for the asserted asymptotic expansion. No numerical regression enters the argument. This proves Theorem A.1.

Appendix B Computational Artifacts

All listed artifacts are available in the [RIME repository](#) under `experiments/paper26/`; short paths below are relative to that directory.

Artifact	Role	Short path
implementation	pair transfer, absorption, Perron gap, and reset length	<code>pair_chain.py</code>
producer	Černý and rare-run finite records	<code>generate_family_results.py</code>
family records	exact fields and bounded float64 spectral observations	<code>results/</code>
analytic supplement	explicit constants and root-window verification	<code>u3-proof.md</code>
partial formalization	TA-III arithmetic and conditional certificate chain	<code>lean/</code>
claim and closure	attribution, evidence status, and exact	<code>claim-surface-map.json</code> ;
metadata	release inventory	<code>release-manifest.json</code>
validation	source digests, finite replay, and Lean closure	<code>validation/</code>

The JSON records reproduce finite instances of the exact family formulas; they do not prove the asymptotic statements. The explicit U3 document is supplementary verification for Theorem A.1. A passing package validator establishes local path, status, hash, and replay closure. It is not independent mathematical validation and does not replace the manuscript proofs.

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